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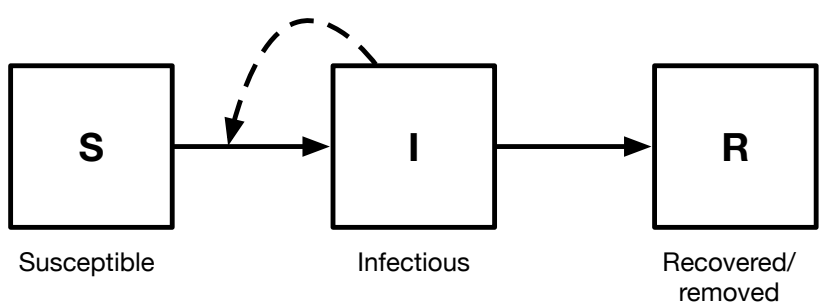
Lab5: Epidemics
Due by Monday May 4th, 2026 @ 11:59pm

1 Outbreaks: From Deterministic Models to Stochastic Realizations

How do public health scientists use models of outbreaks to forecast the shape of outbreaks to come, to interpret the shape of outbreaks that have already taken place, and/or to determine how best to intervene? This laboratory provides a direct path towards building dynamic simulation approaches to all of these questions. In doing so, the laboratory will leverage the core ideas of last week's lectures to transform epidemic theory into simulation dynamics. The focus is on the Ebola virus disease outbreak in West Africa from 2014-2015 as the basis for the choice of model parameters, connecting models with real-world data and ongoing issues of control. The issues of how to connect population models is ongoing; there have been multiple EVD outbreaks since then, including one in 2018 in the Democratic Republic of Congo, with some concern that these epidemics will become endemic. Nonetheless, there is also optimism, given the development of the 'rVSV-ZEBOV' vaccine that protects against the Zaire strain (the same strain that was the causative agent of the 2014-2015 outbreak).

It is also critical to keep in mind that variants of the kinds of models developed in this laboratory are used, in practice, by the CDC, WHO, and others to help policy makers decide how best to allocate resources, design vaccination programs, public health campaigns, and other interventions to prevent and control disease. Indeed, one could argue that despite their simplicity, this laboratory actually goes above and beyond some intervention-level tools. For example, the initial 'Ebola Response Modeling tool' developed by the CDC to provide real-time feedback and scenario response was not a program written in Python, MATLAB, C, but instead a Microsoft Excel spreadsheet. The tool was announced via the CDC `https://www.cdc.gov/media/releases/2014/s0923-ebola-model-Factsheet.html` and is available for inspection here: `https://stacks.cdc.gov/view/cdc/24900`. You might wonder: why an Excel spreadsheet? Answering such a question would take time – more time than we have today – but suffice it to say that the kinds of models developed here have greater flexibility but do require that one knows how to program in order to manipulate them. From a scientific perspective, there is a deeper advantage, the core equations in the CDC Modeling Tool include feedback between different types of individuals, whether susceptible, infectious, recovered, or removed, as well as potential interventions to move individuals from one state to another (e.g., via safe burial) or to change interaction rates between individuals (e.g., via quarantine). Yet, by using a high-level programming language and not a spreadsheet, we are able to develop a suite of flexible approaches including dynamic and stochastic models. These models are also more robust – technically – then a spreadsheet whose update rules by default must involve a Euler-like integration scheme.



In brief, this week's lab will teach you how to (i) model a SIR epidemic outbreak; (ii) explore the relationship between speed, strength, and size of an outbreak; (iii) develop a stochastic simulation of disease dynamics, flexible enough to be applied to real-world cases. This is a lot - but it builds on concepts you've already explored earlier in the class. As always, you can do it!

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2 Epidemic Model - Fundamentals

2.1 Transitions to Outbreaks

Consider an infectious diseases that can spread in a population of size N. Individuals may be susceptible, infectious, or recovered/removed. We term these individuals S, I and R respectively, such that $S+I+R=N$. As was discussed in class, the re-scaled dynamics can be written as

$$\begin{aligned}\dot{S} &= -\beta SI \\ \dot{I} &= \beta SI - \gamma I \\ \dot{R} &= \gamma I\end{aligned}$$

given the two processes, infection at a rate βSI and recovery at a rate γI. Here, this model denotes the fraction of individuals of each type, i.e., $S+I+R=1$. Simulations of this model should use the following core code, beginning with the representation of the ODE:

```
function dydt = sir_model(t,y,pars)
% function dydt = sir_model(t,y,pars)
% SIR Model

S=y(1);
I=y(2);

% The model
dSdt = -pars.beta*S*I;
dIdt = pars.beta*S*I-pars.gamma*I;
dRdt = pars.gamma*I;

dydt = [dSdt; dIdt; dRdt];
```

Applying this model requires a relevant choice of epidemiological parameters. Consider susceptible individuals that interact with c individuals per unit time, of which I/N are infectious, and of which a fraction p of such contacts lead to a new infetious event. In that case we expect the rate of transmission per suseceptible invididual to be cpI/N multiplied by the numnber of susecptible individuals to yield $cpSI/N$ or $\beta \equiv cp$.

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Exercises 1 (25 pts): Complete the simulation code below for the spread of an infectious disease beginning with 1 individual out of 10000 and estimate the value of R0. And run the code. Do you expect the disease to spread or not? Then, change the value of p to 0.01 – will the disease spread, why or why not? (expecting two plots)

```
% main data goes here
pars.c = 20;    % Contacts per unit time (e.g., days)
pars.p = 0.025;         % Probability of infectious contact
pars.beta = ...  % Transmission rate
pars.gamma = 1/4; % Recovery rate (days^-1)
pars.R0 = ...
pars.N = 10000;
pars.I0= 1;
pars.S0= pars.N-pars.I0;

% Run the model
[t,y]=ode45(@(t,y) sir_model(t,y,pars),[0 100],[pars.S0 pars.I0 0]/pars.N);

% Plot the results
tmph=plot(t,y);
set(tmph,'linewidth',3);
xlabel('Time (days)');
ylabel('Population fraction');
tmplh = legend('Susceptible','Infectious','Recovered');
legend('boxoff');
```

2.2 Speed, Strength, and Size

Once an outbreak starts, how far will it go? That is to say: how many individuals in a population will become sick? In class, we discussed the relationship between speed, strength, and size. The speed can be measured in terms of the rate, r, of exponential growth of the outbreak, i.e., $I(t) \sim e^{rt}$. The strength can be measured in terms of the dimensionless number $\mathcal{R}_0$. Finally, the size of the outbreak can be measured in terms of the fraction of the population infected, $R(t \to \infty)$, which is equivalent to $1 - S(t \to \infty)$. In class we derived the strength-size relationship, specifically:

$$\mathcal{R}_0(S_\infty - 1) = \log(S_\infty) \tag{1}$$

as well as the strength-speed relationship, i.e.,

$$r = \gamma(\mathcal{R}_0 - 1) \tag{2}$$

Here, you will explore both of these sytematically, gaining an intuition for these two relationships.

2.2.1 Strength and Speed

It would seem that outbreaks with higher values of $\mathcal{R}_0$ would lead to a faster increase in case counts. But that is not necessarily the case. One way to explore this is to modulate both β and γ, i.e., the transmission and recovery rates, which influence both strength and speed. To begin, let's first measure the speed in an outbreak. The code below simulates an outbreak using $\mathcal{R}_0 = 2$, $\beta = 0.5$ days^{-1}, and $\gamma = 0.25$ days^{-1}. For this combination, we expect $r = 0.25$ days^{-1}. The following code illustrates the following steps. First, simulate the model. Second, fit a line to the log-transformed infectious counts, $I(t)$. Note that you could also find the line of best fit to the cumulative case incidence, $I(t) + R(t)$. The slope of this line should be the estimated $\hat{r}$.

```
% Run the model over 10 days
```

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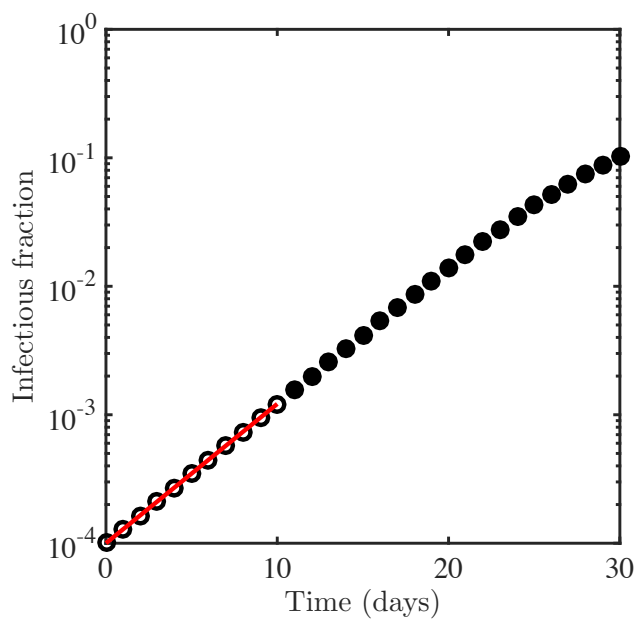
```
[t,y]=ode45(@(t,y) sir_model(t,y,pars),[0:1:10],[pars.S0 pars.I0 0]/pars.N);

% Find the slope
[p,s]=polyfit(t,log(y(:,2)),1);

% Plot the data and overlay the best-fit exponential
tmph=semilogy(t,y(:,2),'ko');
set(tmph,'linewidth',3);
hold on
tmph=semilogy(t,exp(p(1)*t+p(2)),'r-');
set(tmph,'linewidth',3);

% Use solid points for the future k
[t,y]=ode45(@(t,y) sir_model(t,y,pars),[0:1:30],[pars.S0 pars.I0 0]/pars.N);
tmpi=find(t>10);
tmph=semilogy(t(tmpi),y(tmpi,2),'ko');
set(tmph,'linewidth',3,'markerfacecolor','k');
```

When implemented, this code results in the following image:



In this case, the slope of the line is 0.2496 days^{-1} and the expected value is $r = 0.25$ days^{-1}. With this code snippent in mind, your next challenge is to examine the speed-size relationship given variation in β and γ.

Exercises 2 (25 pts): Determine the strength and speed for the following sets of disease parameters. Calculate expected speed and then compare it with the estimated speed (estimated slope of $log(I)$ versus time) and see if they match. Should create 4 plots (similar to the plot above) with corresponding transmission and recovery values.

Transmission β	Recovery γ	Strength $\mathcal{R}_0$	Speed r
0.5	0.4	...	...
1	0.5	...	...
0.25	0.5	...	...
0.75	0.25	...	...

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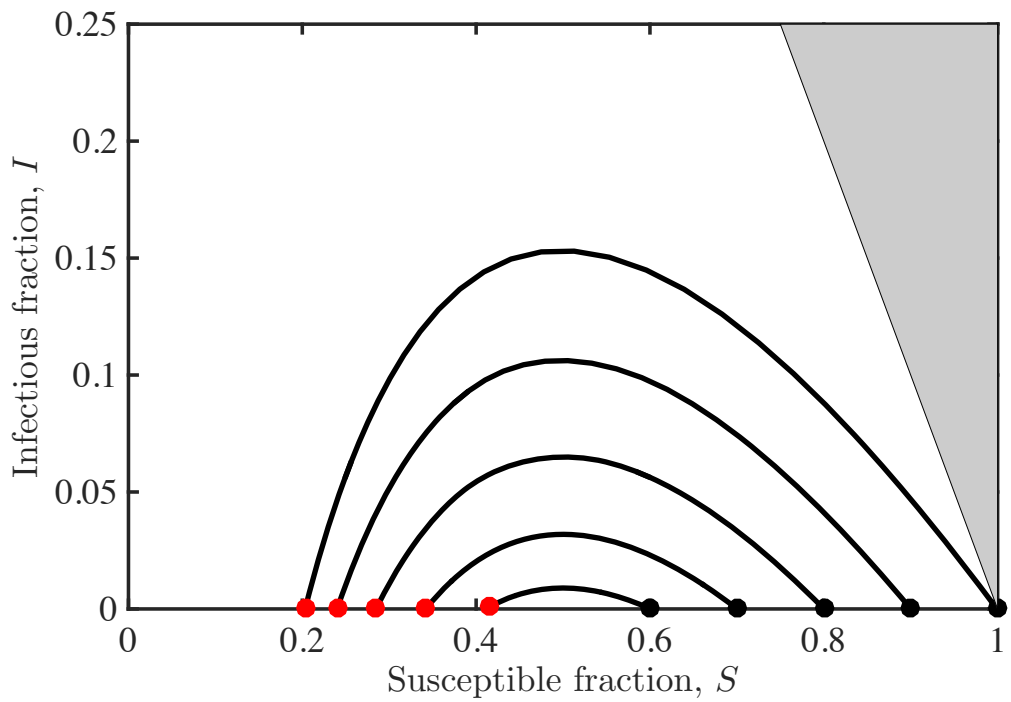
2.2.2 Strength and Size

As should be apparent from the first figure, an outbreak does not reach all. The final size of the epidemic can be measured in terms of the cumulative case count, i.e., the value of R_∞ or as $1 - S_\infty$. The next code set shows how to systematically modify the initial fraction of infected. In these overlaid simulations, the initial values are marked by black circles and the final points by red circles. As should be apparent, there is a critical value of $1/\mathcal{R}_0$ where an initial outbreak will not spread – this is also the basis for findings that not all must be vaccinated for a vaccine to be effective!

```
% Modify the initial values
pars.N = 10000;
pars.S0_range=[0.6 0.7 0.8 0.9 0.999];
pars.I0_range=1/pars.N*ones(1,5);
pars.R0_range=1-pars.S0_range-pars.I0_range;

% Run the model
for i=1:length(pars.S0_range),
  [t,y]=ode45(@(t,y) sir_model(t,y,pars),[0:1:200],[pars.S0_range(i) pars.I0_range(i) pars.R0_range(i)]);
  tmph=plot(y(:,1),y(:,2),'k-');
  set(tmph,'linewidth',3);
  hold on
  tmph=plot(y(end,1),y(end,2),'ro');
  set(tmph,'markerfacecolor','r');
  tmph=plot(y(1,1),y(1,2),'ko');
  set(tmph,'markerfacecolor','k');
end
% Show the excluded regime
patch([1 1 0.75 1],[0 0.25 0.25 0],[0.8 0.8 0.8]);
```

Running this code yields the following trajectories where the shaded region denotes inaccessible parts of phase space given that $S + I \leq 1$.



2.3 From Outbreaks to Endemics

The SIR model, with rapid loss of immunity can be written in terms of an SI model:

$$\dot{S} = -\beta SI + \gamma I$$
$$\dot{I} = \beta SI - \gamma I$$

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Because those that recover are immediately susceptible, there is a new long-term possibility: an endemic disease state where $S^* = \frac{1}{\mathcal{R}_0}$ and $I^* = \frac{\mathcal{R}_0 - 1}{\mathcal{R}_0}$. Hence, the 'SI' model can be coded as follows:

```
function dydt = si_model(t,y,pars)
% function dydt = si_model(t,y,pars)
% SI Model

S=y(1);
I=y(2);

% The model
dSdt = -pars.beta*S*I+pars.gamma*I;
dIdt = pars.beta*S*I-pars.gamma*I;

dydt = [dSdt; dIdt];
```

As should be apparent, there is only an endemice state with $I^* > 0$ when $\mathcal{R}_0 > 1$.

Exercises 3 (25 pts): Given parameters $\beta = 0.3$ and $\gamma = 0.25$ days^{-1}. Simulate and compare the results of SIR and SI model (should have two plots). For these parameters, simulate the dynamics over a 1 year = 365 day period.

3 Stochastic Epidemics

3.1 Definition of the Model

Stochastic realizations of the SIR model are simulated using the Gillespie framework, given the "reaction" events in the following table:

Process	**Reaction**	**rate/probability**
Infection	$S + I \to 2I$	$r_1 = \beta SI/N$
End of infectiousness (recovery)	$I \to R$	$r_2 = \gamma I$

These processes denote transitions amongst individuals who are susceptible (S), infectious (I), and recovered/removed (R). The total population is fixed at $N = S + I + R$. Epidemics are initiated with one infectious individual in an otherwise susceptible population. Mathematically, the initial state is, in the respective ordering, $\mathbf{y} = (N_0 - 1, 1, 0)$ at $t = 0$. The total rate of outbreak-associated events is $r_{tot} = \sum_{i=1}^{2} r_i$. The time until next event is determined randomly such that $\delta t \sim \frac{-\log \chi}{r_{tot}}$ where χ is a uniformly distributed number between 0 and 1. In this way, the time between events follows an exponential distribution with rate r_{tot}. Then, the probability of each event is r_i/r_{tot}. After selecting an event and updating the discrete number of individuals, the reaction rates are recalculated and the process continues. The same framework can be extended to include other classes (e.g., exposed individuals) as well as other kinds of processes (e.g., post-death transmission in the case of EVD). Trajectories are complete when the epidemic dies out because there are no more infectious individuals. In the present context, we are interested in those trajectories that do not die out before the end of the simulation time.

3.2 Simulating a stochastic outbreak

Here, your objective is two-fold, first to modify the code below to simulate a stochastic epidemic, and then to simulate and compare results from stochastic simulations to your deterministic trajectories.

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Exercises 4 (25 pts): Modify the following code to implement a stochastic SIR model including both transmission and recovery events in a population of size N. For the code, you can just include the lines you modified in your solution. After the modification, simulate the system and compare it with the deterministic SIR model (include a figure similar to the one below).

```
function [t,y] = stochsim_SIR(trange,y0,pars)
% function [t,y] = stochsim_SIR(trange,y0,pars)
%
% Simulates an SIR model via the Gillespie algorithm
% from t0 to tf in trange given initial
% conditions in y0 = [S0 I0 R0] and parameters
% in pars.  Returns time and values

% Conditions
t0=trange(1);
tf=trange(2);
t(1)=t0;
y(1,:)=y0;
tcur=t0;
ycur=y0;
ind=1;

% Model
while (tcur<tf)
  % Check to see if there is an infection
  if (ycur(2)==0)
    ind=ind+1;
    t(ind)=tf;
    y(ind,:)=ycur;
    break;
  end
  % Rates
  infrate = ...
  recrate = ...
  totrate = infrate + recrate;
  dt = -1/totrate*log(rand);
  tcur=tcur+dt;
  % Event type
  if (rand<(infrate/totrate)) % infection
    ...
    ...
  else % recovery
    ...
    ...
  end
  ind=ind+1;
  t(ind)=tcur;
  y(ind,:)=ycur;
end
```

Once you have a working code, simulate it using the following parameters including an initial seed of 10 infectious individuals. This is usually enough to ensure the disease spreads. parameters

```
pars.c = 20;    % Contacts per unit time (e.g., days)
pars.p = 0.025;         % Probability of infectious contact
pars.beta = pars.c*pars.p;  % Transmission rate
pars.gamma = 1/4; %
pars.tf=60;
pars.R0 = pars.beta/pars.gamma;
pars.N = 1000;
pars.I0= 10;
pars.S0= pars.N-pars.I0;
```

and compare directly to the SIR model which should yield the following kind of result

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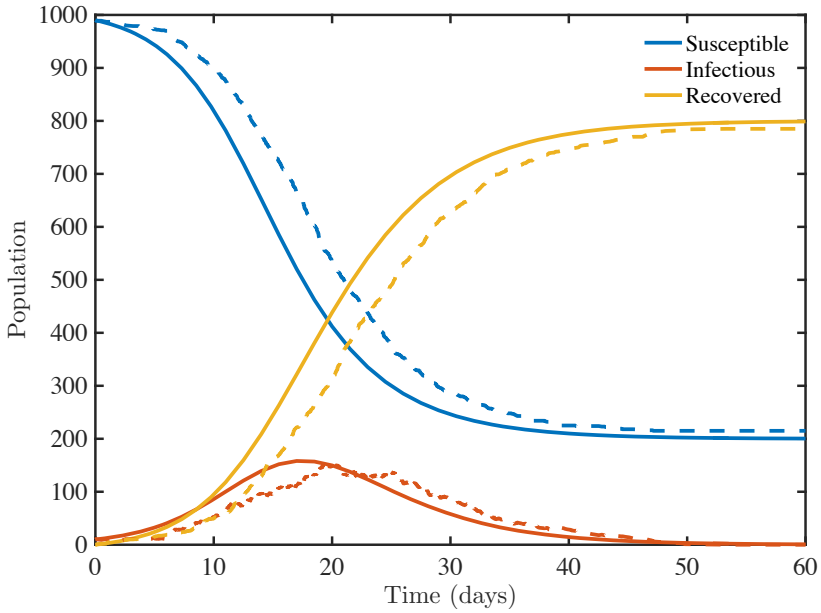

As is apparent, there can be many differences between a deterministic model and a stochastic trajectory. Understanding that difference is at the very heart of inferring and predicting disease trajectories in real systems. If you have time, you could even try to estimate the speed of the outbreak for a set of trajectories... you will find that there is variability even without considering observational noise or process noise, and that such variation can be substantial at the ouset of an outbreak when estimates are most needed!

Finally, it may seem non-trivial to copmare the models, but it is easier than it seems. The key is to run both models but then recall that the ODE model dynamics are in terms of fractions, which must be re-scaled. Here is how to do it:

```
% Run the ODE model
[t,y]=ode45(@(t,y) sir_model(t,y,pars),[0 pars.tf],[pars.S0 pars.I0 0]/pars.N);
tmph=plot(t,pars.N*y);
set(tmph,'linewidth',3);
hold on

% Run the stochastic model
[tsim,ysim]=stochsim_SIR([0 pars.tf],[pars.S0 pars.I0 0],pars);
tmphsim=plot(tsim,ysim,'--');
set(tmphsim,'linewidth',3);

% Color the lines the same way (where dashes are for stochastic)
for i=1:3,
  set(tmphsim(i),'color',get(tmph(i),'color'));
end
```

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